STACKS, SHEAVES ON ALGEBRAIC STACKS: LEMMA-PUSHFORWARD-RESTRICTION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4091-4110 of sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex; proof span is lines 4112-4141, the complete author proof (28 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 24 of queue_algebraic-stacks-sheaves.json (pb=1159, stmt 4091-4110, proof 4112-4141).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-pushforward-restriction' also exists in stacks-cohomology — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 04Y5: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 4141–4141. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma equation-pushforward: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 747–751. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-fibre-products: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2864–2873. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-pullback-injective: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 3009–3022. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-pullback-injective-modules: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 3044–3063. Cited

Date: stacks-project:stacks-sheaves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : \mathcal{X} \to \mathcal{Y}$ be a 1-morphism of algebraic stacks[1] over S. Let $\tau \in \{Zariski, étale, smooth, syntomic, fppf\}$. Let $\mathcal{F}$ be an object of $Ab(\mathcal{X}_\tau)$ or $Mod(\mathcal{X}_\tau, \mathcal{O}_\mathcal{X})$. Then the sheaf $R^i f_* \mathcal{F}$ is the sheaf associated to the presheaf*

$$y \longmapsto H^i_\tau\Big((Sch/V)_{fppf} \times_{y,\mathcal{Y}} \mathcal{X},\ pr^{-1}\mathcal{F}\Big)$$

Here y is an object of $\mathcal{Y}$ lying over the scheme V.

Proof. Choose an injective resolution $\mathcal{F}[0] \to \mathcal{I}^\bullet$. By the formula for pushforward (equation-pushforward) we see that $R^i f_* \mathcal{F}$ is the sheaf associated to the presheaf which associates to y the cohomology of the complex

$$\begin{array}{c} \Gamma\Big((Sch/V)_{fppf} \times_{y,\mathcal{Y}} \mathcal{X},\ \mathrm{pr}^{-1}\mathcal{I}^{i-1}\Big) \\ \downarrow \\ \Gamma\Big((Sch/V)_{fppf} \times_{y,\mathcal{Y}} \mathcal{X},\ \mathrm{pr}^{-1}\mathcal{I}^{i}\Big) \\ \downarrow \\ \Gamma\Big((Sch/V)_{fppf} \times_{y,\mathcal{Y}} \mathcal{X},\ \mathrm{pr}^{-1}\mathcal{I}^{i+1}\Big) \end{array}$$

Since pr^{-1} is exact, it suffices to show that pr^{-1} preserves injectives. This follows from Lemmas lemma-pullback-injective and lemma-pullback-injective-modules as well as the fact that pr is a representable morphism of algebraic stacks (so that pr is faithful by Algebraic Stacks, Lemma 04Y5 and that $(Sch/V)_{fppf} \times_{y,\mathcal{Y}} \mathcal{X}$ has equalizers by Lemma lemma-fibre-products). □

[1] This result should hold for any 1-morphism of categories fibred in groupoids over $(Sch/S)_{fppf}$.